

Landmark embeddings of multi-copy gene family trees

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Abstract

Phylogenomic data sets contain gene trees that differ because of biological processes, sampling, and inference error. Direct comparison is particularly difficult when gene families contain paralogs, differ in taxon coverage, or have few species in common. We introduce a landmark representation in which each gene family tree is mapped to a fixed-length vector of dissimilarities to a common collection of reference species trees. The dissimilarities describe complementary aspects of gene–species tree disagreement, including duplication, loss, deep coalescence, Robinson–Foulds, split-matching, and approximate subtree-prune-and-regraft scores. Once the reference trees are fixed, new gene trees can be projected independently without recomputing an all-pairs distance matrix. Simulations with missing taxa and multi-copy gene trees, together with an exploratory fungal data set, illustrate how these vectors can be visualised and clustered. The present analyses establish a proof of concept: replicated benchmarking, explicit treatment of gene-tree uncertainty, and biological validation of the empirical clusters remain necessary before comparative performance or biological conclusions can be claimed.

1 Introduction

A gene tree is an estimate of the evolutionary history of a genomic locus or gene family. Gene trees need not agree with one another or with a species tree: incomplete lineage sorting, duplication, loss, horizontal transfer, hybridisation, recombination, taxon sampling, model

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misspecification, and finite sequence information can all contribute to discordance. The most general input is therefore a multi-copy gene family tree, in which several leaves can map to the same species and some species can be absent.

Collections of gene trees are used both to infer species histories and to investigate heterogeneity across the genome. Coalescent summary methods such as ASTRAL and ASTRID were originally formulated for single-copy inputs (Mirarab and Warnow, 2015; Vachaspati and Warnow, 2015). More recent methods explicitly use multi-copy gene trees. ASTRAL-Pro defines paralogy-aware quartet similarity (Zhang et al., 2020), whereas SpeciesRax estimates rooted species trees under a duplication–transfer–loss model (Morel et al., 2022). These methods demonstrate that paralogs need not be discarded, but their primary objective is species-tree inference. Our objective is complementary: to construct a reusable coordinate representation of each gene family tree for exploratory comparison, visualisation, clustering, or supervised learning.

Many tree-set visualisation methods begin with pairwise distances between all trees and then use multidimensional scaling, spectral methods, or related projections (Gori et al., 2016; Jombart et al., 2017; Huang et al., 2016). Other work maps trees directly to vectors of splits, shape descriptors, or other features (Leigh et al., 2008, 2011; Susko et al., 2006; Narechania et al., 2016; Nye, 2011; Yoshida et al., 2015; Lewitus and Morlon, 2015; Kendall and Colijn, 2016; Colijn and Plazzotta, 2018). Quartet concordance factors and simplex plots provide a further model-based summary when gene trees share taxa (Allman et al., 2022). These approaches are valuable, but an all-pairs dissimilarity matrix must be extended when new trees arrive, and many standard distances require compatible leaf labels. Both issues become acute for large collections of multi-copy families with heterogeneous taxon coverage.

We instead represent a gene tree by its dissimilarities to a fixed collection of landmark trees. The landmarks are rooted candidate species trees containing the full species set. This construction is related to landmark-based manifold representations and centroid methods (Jeon et al., 2001; Park et al., 2003; Chen and Cai, 2011; Vin De Silva, 2003): after choosing the landmarks, any gene tree can be projected independently. More than one dissimilarity can be evaluated against each landmark, so distinct summaries of duplication, loss, coalescence, and topological disagreement form complementary views of the same gene tree.

The resulting vector is called a *tree signature*. Two gene trees with similar signatures make similar patterns of agreement and disagreement with the reference trees under the selected

measures. This does not imply that their histories are identical, nor does the representation by itself identify the biological cause of discordance. It provides a common feature space in which downstream statistical methods can be applied and in which individual features remain associated with explicit reference trees and dissimilarities.

This fixed-coordinate property distinguishes the proposal from an embedding derived only from the sample's pairwise distance matrix. It also creates an important dependence: the geometry is conditional on the selected reference trees. Reference selection is therefore part of the model rather than a neutral preprocessing step, and its stability must be evaluated when signatures are used out of sample.

2 Methods

2.1 Inputs and tree signatures

Let G_j be an unrooted gene tree whose leaves are gene copies, each mapped to a species label. Multiple leaves may map to the same species, and species may be absent. Let $\mathcal{R} = \{R_1, \dots, R_L\}$ be an ordered collection of rooted reference species-tree topologies. Every reference tree contains the same full species set, which must include every species represented in G_j . Reference branch lengths are not used by the current dissimilarities.

For an ordered collection of $K = 6$ dissimilarities, the unnormalised tree signature is

$$\mathbf{v}(G_j; \mathcal{R}) = (d_i(G_j, R_\ell))_{\ell=1, \dots, L; i=1, \dots, K}. \quad (1)$$

The implementation orders the six values for each reference tree as duplication, loss, deep coalescence, Robinson–Foulds (RF), Hdist, and approximate subtree-prune-and-regraft (SPR). The resulting vector has $6L$ coordinates. Reference topologies that are identical under the parser's rooted-tree convention are stored once, so L is the number of distinct parsed reference topologies.

The native Python interface represents parsed Newick data as reusable `TreeSpace` objects. A call uses the first distinct gene topology and every distinct reference topology. Thus, a posterior sample or bootstrap distribution is currently represented by projecting its trees separately; the software does not yet define a family-level aggregation of multiple topologies. The parsed

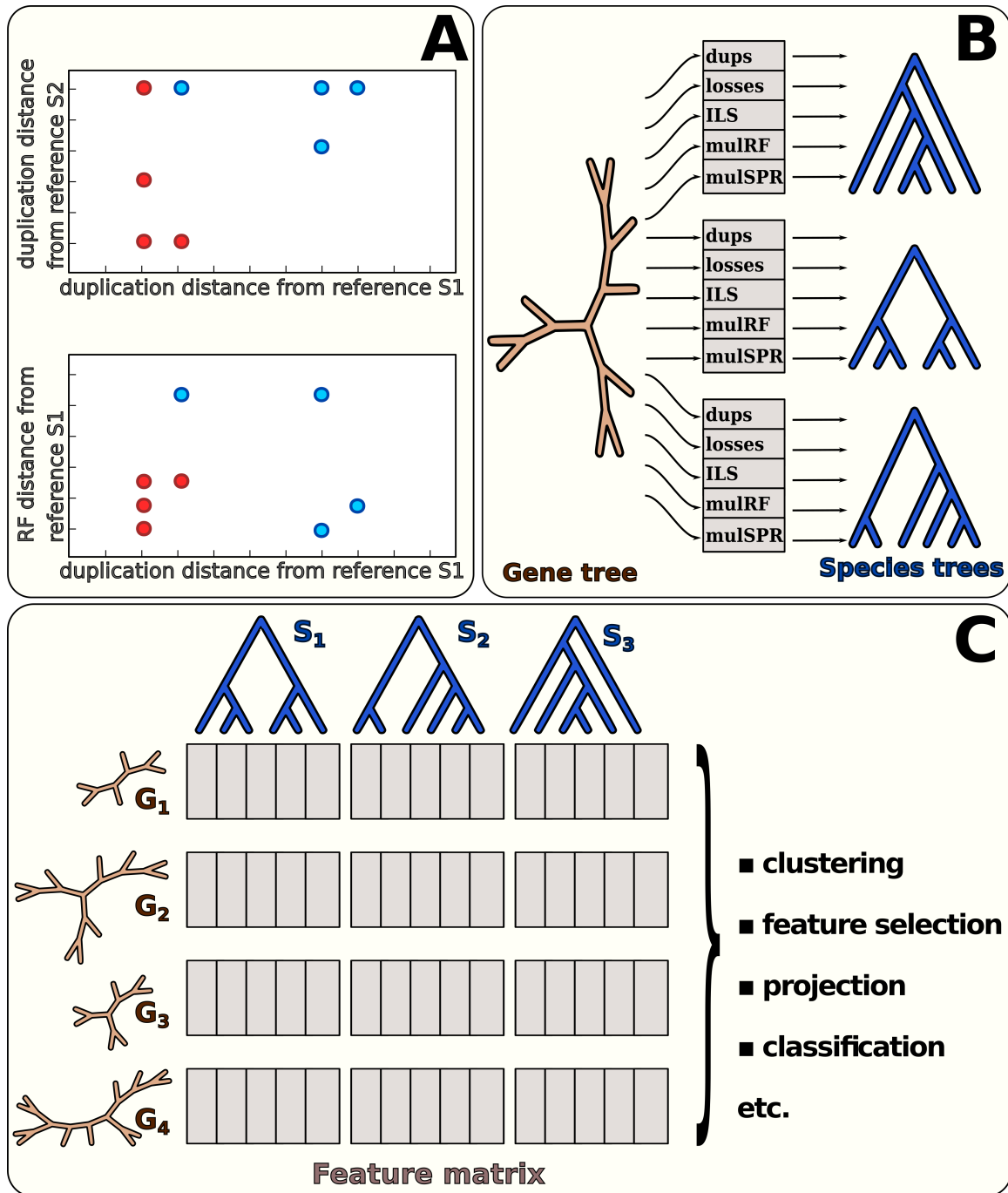


Figure 1: Schematic representation of a tree-signature calculation. In panel A, gene trees are compared either with two reference species trees using one dissimilarity, or with one reference species tree using two dissimilarities. Either view alone has limited resolution, whereas their combination distinguishes the illustrated groups. Panel B shows the signature of one gene tree across combinations of reference trees and dissimilarities. The current implementation treats gene trees as unrooted and reference species trees as rooted, although some dissimilarities ignore the reference root. Panel C shows the resulting design matrix used by downstream analyses.

objects are retained across calls, while calculation-specific copies protect them from algorithms that temporarily alter reconciliation or topology state.

2.2 Dissimilarities

Three reconciliation scores are inherited from the LCA-mapping calculations used by *guenomu* (De Oliveira Martins et al., 2016): the minimum number of duplications, losses, and deep coalescences. Gene-tree root position is not assumed to be known. The implementation evaluates possible gene roots and retains the minimum reconciliation score; the reference species tree remains rooted.

The remaining scores use an extended species tree, following the construction used by mulRF (Chaudhary et al., 2015). Each species leaf is replaced by a multifurcation containing the gene-copy leaves mapped to that species. The extended species tree and the gene tree then have a one-to-one leaf mapping. RF is the unrooted Robinson–Foulds distance between these trees. Hdist sums split-matching costs obtained through a minimum-cost assignment, and the SPR score is an approximation based on the same split matching. These three calculations ignore the reference root. All six scores use topology but not branch lengths.

Although we use the convenient term *dissimilarity*, the gene–species scores are not metrics on a common tree space: their arguments have different roles, and symmetry and the triangle inequality are not generally applicable.

2.3 Normalisation

The software exposes three output modes. The unnormalised mode returns Equation 1. The default rescaled mode divides each score by an implementation-defined upper bound $u_i(G_j)$ based on gene-tree size, species coverage, and the corresponding split representation:

$$s_i(G_j, R_\ell) = \frac{d_i(G_j, R_\ell)}{u_i(G_j)}. \quad (2)$$

This mode returns $6L$ values and requires no randomisation.

An optional randomisation mode returns two consecutive $6L$ blocks. For each observed score it first computes an empirical lower-tail rank against the other observed reference scores and

Q randomised gene–species comparisons:

$$p_i(G_j, R_\ell) = \frac{\sum_{r \neq \ell} I[d_i(G_j, R_r) \leq d_i(G_j, R_\ell)] + \sum_{q=1}^Q I[\tilde{d}_{ijq} \leq d_i(G_j, R_\ell)]}{L - 1 + Q}, \quad (3)$$

where $I[\cdot]$ is the indicator function and $\tilde{d}_{ijq}$ is obtained after randomising both the working gene and species topology. The second block rescales the observed score using the minimum and maximum across the observed and randomised comparisons for gene tree G_j :

$$m_i(G_j, R_\ell) = \frac{d_i(G_j, R_\ell) - a_{ij}}{b_{ij} - a_{ij}}. \quad (4)$$

When $a_{ij} = b_{ij}$ within numerical tolerance, the implementation returns one for that coordinate. We call Equation 3 an empirical rank rather than a calibrated p -value because the reference trees and randomised comparisons do not necessarily constitute exchangeable draws from a specified null distribution.

2.4 Choice of reference trees

For N labelled species there are $(2N - 3)!!$ rooted binary topologies, already more than 10^7 for $N = 10$. Exhaustive use is neither possible nor desirable. The reference collection should instead span alternatives relevant to the observed gene families and the scientific question. Candidate sources include posterior or bootstrap samples of species trees, optima inferred from subsets of genes, deliberate perturbations around such trees, and random trees used as negative controls.

A small reference set reduces dimension and computation, but can map distinct gene histories to similar signatures. A large or redundant set increases cost without necessarily adding information. Because the resulting geometry is conditional on $\mathcal{R}$, reference selection should be assessed by sensitivity analyses: repeat the downstream analysis across independently selected reference sets, compare informed and random landmarks, and reserve gene families not used to construct $\mathcal{R}$ when evaluating out-of-sample behaviour.

2.5 Downstream analyses

Tree signatures form a conventional design matrix with one row per projected gene tree and one column per reference-tree–dissimilarity combination. Standard dimension-reduction, clustering, and supervised-learning methods can therefore be applied. These downstream procedures do not turn the coordinates into independent observations, and a visual cluster is not by itself evidence for a distinct biological process. Feature loadings can nevertheless be traced back to particular dissimilarities and reference trees.

3 Results

We evaluated the representation in two simulation scenarios and one archived empirical example. These analyses were designed as demonstrations of the geometry of tree signatures. They were not replicated sufficiently to support formal comparisons between algorithms, and the empirical example is reported as exploratory pending reconstruction from a versioned data release.

3.1 SPR-chain simulation with missing taxa

Starting from a random tree on 16 leaves, we generated a chain in which consecutive trees differed by an SPR move. Four groups of ten gene trees were sampled from separated portions of the chain. Eight leaves were removed from each sampled tree; trees in the same group shared the same removed leaves. Reference trees retained all leaves. We compared an informed reference collection drawn from portions of the SPR chain with a collection of random reference trees.

Figure 2 shows metric multidimensional-scaling (MDS) projections of the resulting signatures. The four generating groups are visually separated with the informed landmarks. Separation is also apparent with random landmarks, although the geometry changes. Because taxon-removal patterns were shared within groups, this demonstration does not distinguish phylogenetic signal from signal introduced by missingness. No accuracy statistic or uncertainty interval was recorded for the archived run.

The trees in this scenario are single-copy and can therefore also be compared by conventional pairwise distances. An archived exploratory comparison used normalised RF and approximate SPR pairwise distances and produced qualitatively similar MDS plots; those unrecorded plots are not treated as evidence here.

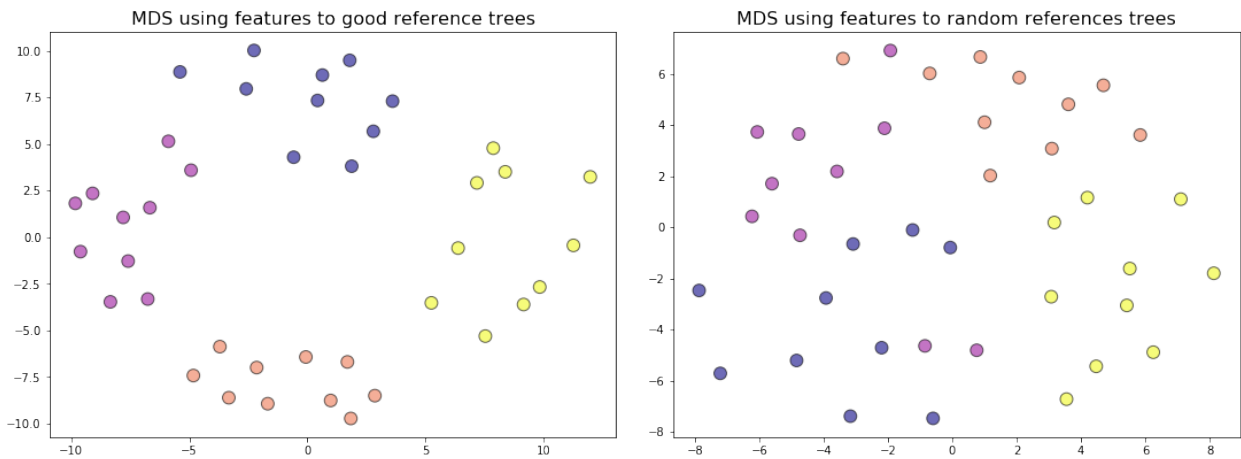


Figure 2: SPR-chain simulation with missing taxa. MDS projections use signatures calculated from informed reference trees drawn from the SPR chain (left) or random reference trees (right). Colours denote four groups sampled from separated portions of the chain. The plot is one exploratory realisation; points do not represent independent simulation replicates.

3.2 Genome-level simulation

The second scenario used SimPhy (Mallo et al., 2015) to generate gene trees within locus trees evolving inside species trees. The archived notebook generated four species trees with 20 species and 50 gene families per species tree, sampled one or two individuals per species, and removed up to approximately half of the gene-tree leaves. Reference trees were generated by applying SPR perturbations to the four species trees; the true species trees themselves were excluded from the signature basis.

The archived parameterisation included duplication, loss, transfer, and incomplete lineage sorting. Figure 3 shows the family-size and taxon-overlap distributions and four two-dimensional projections from one run. MDS and Isomap show visible separation among the four generating species trees. The t-SNE projection overlaps strongly, and the spectral embedding arranges the groups along narrow branches. Thus, the demonstration is sensitive to the projection method and should not be interpreted as a general clustering benchmark.

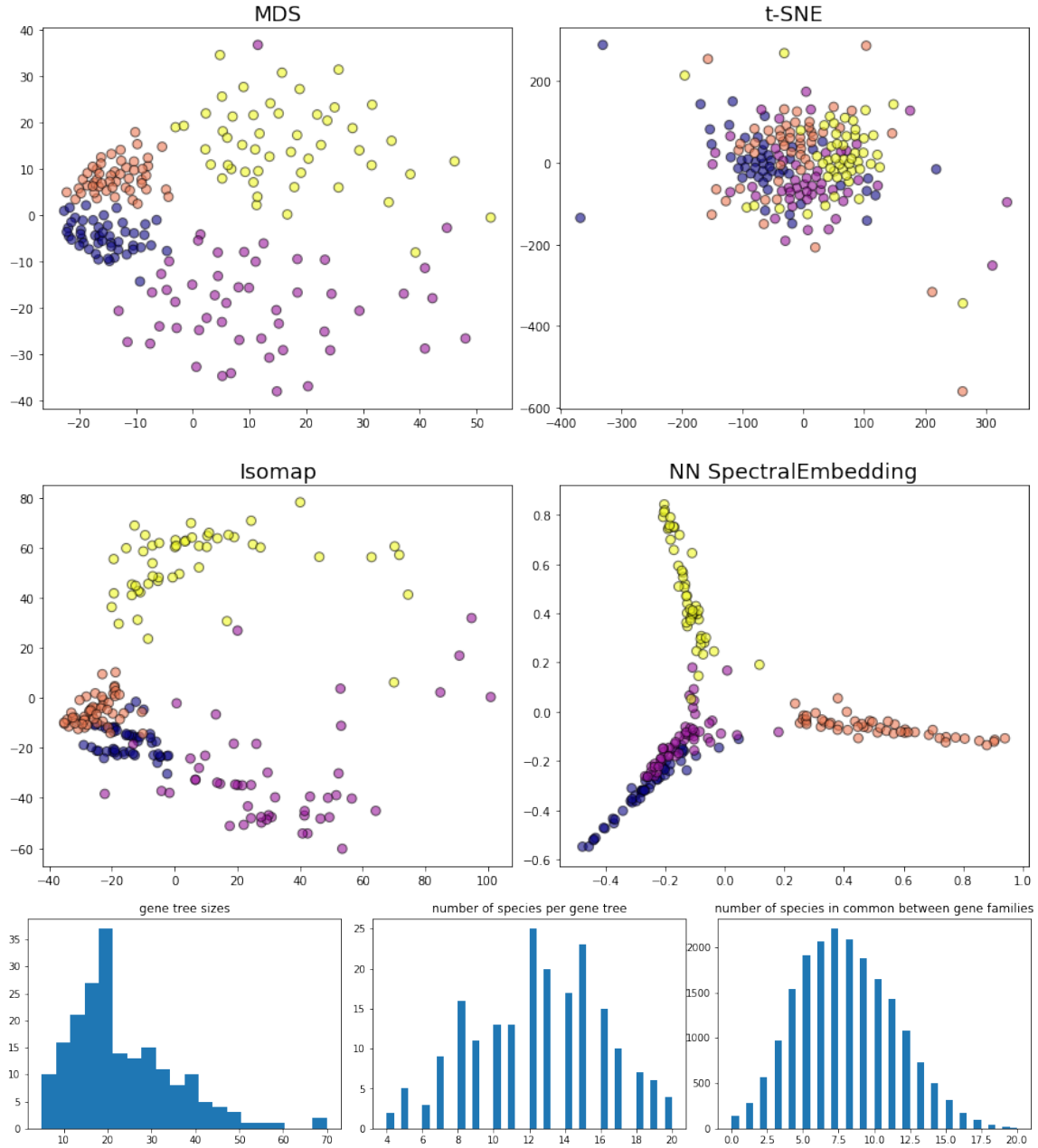


Figure 3: One SimPhy realisation. The upper image compares MDS, t-SNE, Isomap, and nearest-neighbour spectral embeddings of the tree signatures; colours denote the four generating species trees. The lower image summarises gene-tree leaf counts, species coverage, and pairwise species overlap. Axis coordinates in the embeddings have no direct biological units.

3.3 Archived fungal example

The archived fungal analysis used root hierarchical orthologous groups (root HOGs) from an approximately 2018 OMA snapshot. At a specified taxonomic level, a HOG represents orthologs and paralogs descending from a common ancestral gene (Altenhoff et al., 2024). The original analysis records do not identify a frozen OMA release, HOG manifest, taxon list, or complete gene-tree inference pipeline. Approximate counts are retained here only to

document the historical proof of concept: about 11,000 gene families across 49 species, with approximately 50 high-coverage root HOGs and 120 additional large families used to help infer reference trees.

Reference candidates were inferred with *guenomu* (De Oliveira Martins et al., 2016), augmented by topology perturbations, and contrasted with randomised references. The archived notes describe approximately 250 informed and 250 random reference trees. The downstream analysis used the empirical MinMax block and omitted the empirical-rank block because the latter did not appear useful in preliminary projections. No predefined criterion for that decision was recorded.

Sparse principal-component analysis (PCA) selected different mixtures of dissimilarities when using informed and random reference trees (Figures 4 and 5). For the informed references, individual components were often dominated by one or two dissimilarity types. The random-reference components were less readily attributable to a single type. This is a descriptive loading analysis; it does not establish that a selected dissimilarity corresponds to a unique evolutionary process.

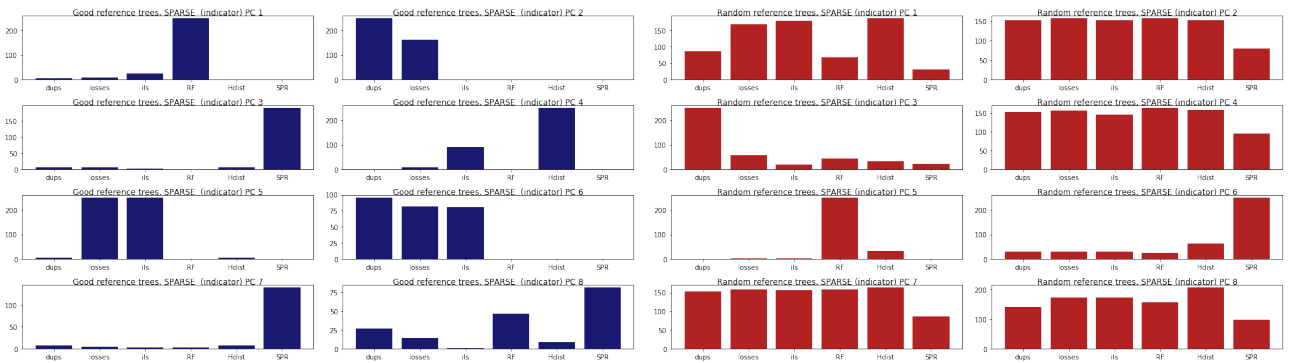


Figure 4: Composition of the first eight sparse principal components for informed reference trees (left) and random reference trees (right). Each bar counts non-zero feature loadings, grouped by dissimilarity. A feature is one reference-tree–dissimilarity combination, not one gene family.

The full signature matrix also produces visible structure under PCA, Isomap, and spectral embedding (Figure 6). The colours in all panels are five-means labels fitted to the full feature matrix; they are not external biological annotations. Their persistence across projections illustrates the geometry imposed by that fitted partition, but does not validate five biological classes or show that informed references outperform random references.

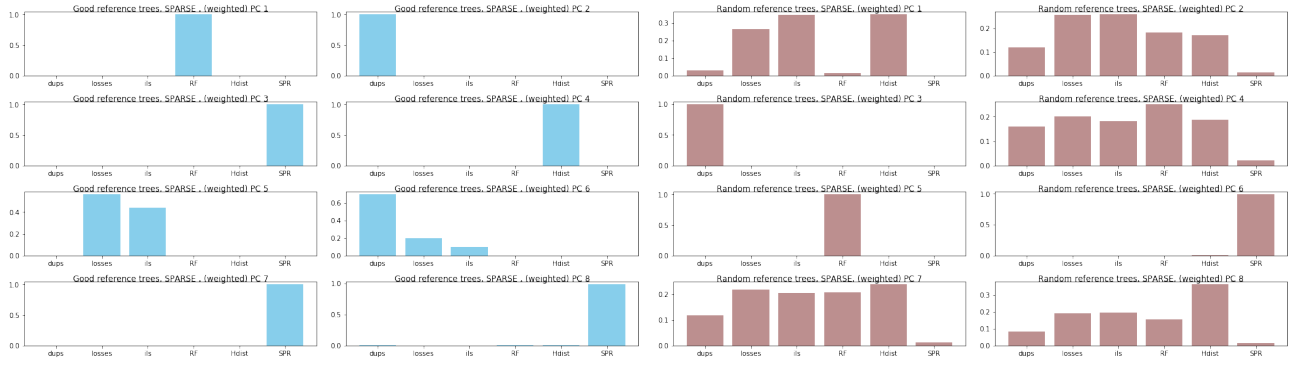


Figure 5: Loading weight of the first eight sparse principal components for informed reference trees (left) and random reference trees (right). Each bar is the sum of squared feature loadings for one dissimilarity type.

3.4 Reproducibility status

The simulations can be reconstructed from notebooks in the software repository, but they currently contain absolute paths, unrecorded random states, and version-dependent analysis calls. The empirical input data and analysis code are not present. Consequently, all results in this section should be considered provisional until the workflows are made portable, parameterised, and repeatable from archived inputs.

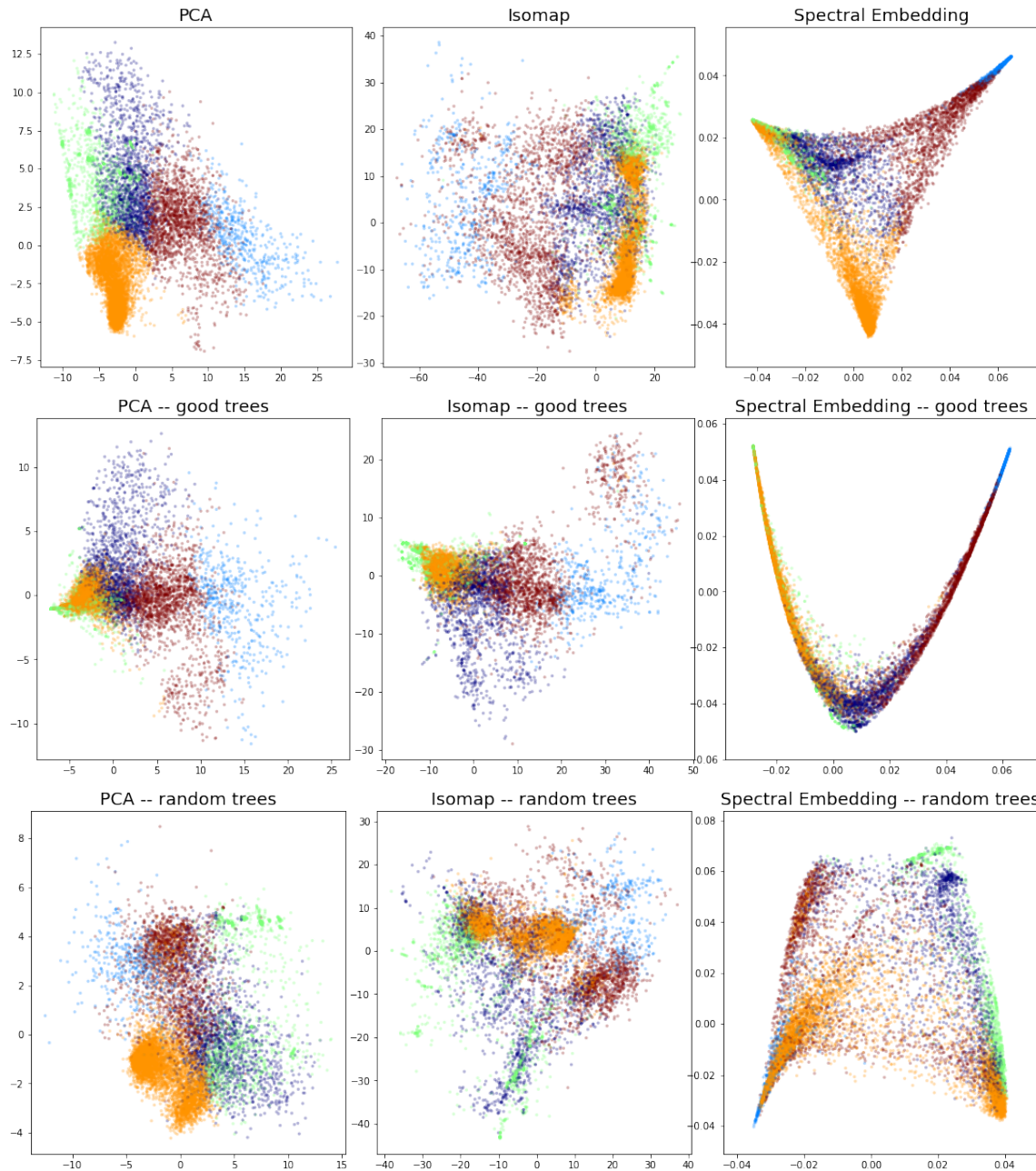


Figure 6: Exploratory projections of archived fungal gene families using all reference trees (top), informed reference trees (middle), or random reference trees (bottom). Columns show PCA, Isomap, and spectral embedding. Colours are five-means labels estimated from the full signature matrix and reused in every panel.

4 Discussion

Tree signatures convert heterogeneous gene family trees into vectors indexed by common reference species trees. Their main practical advantage is a separation between fitting the coordinate system and projecting observations: once the reference collection is fixed, a new gene tree can be evaluated without recomputing distances among all previously observed trees. Unlike an opaque generic embedding, every coordinate retains an interpretation as a particular gene–species tree dissimilarity. The reusable native tree-space interface strengthens this design by avoiding repeated Newick parsing when the same trees are used across calculations.

The representation is process-agnostic only in a limited sense. It does not fit a joint generative model of gene and species evolution, but several coordinates are motivated by reconciliation events. A small duplication score, for example, is evidence of topological compatibility under an LCA mapping; it is not an estimate of a duplication rate or proof that duplication generated the observed discordance. Methods such as ASTRAL-Pro and SpeciesRax instead optimise a species-tree objective that explicitly accommodates multi-copy families (Zhang et al., 2020; Morel et al., 2022). Tree signatures are complementary: they are intended for exploration and prediction across gene families rather than as a replacement for mechanistic species-tree inference.

The reference trees determine the coordinate system and therefore influence distances, projections, and clusters. This dependence is a feature when the landmarks encode hypotheses of interest, but it also creates risks. Landmarks inferred from the same gene families can introduce circularity, and a basis covering one sample may represent future families poorly. Informed and random references produced different loading patterns in the archived analyses, but those examples do not identify an optimal selection rule. Landmark stability, redundancy, and out-of-sample coverage should be treated as model-selection questions.

Several limitations arise from the dissimilarities. The present implementation discards branch lengths, and three scores ignore the reference root. Reconciliation scores depend on the species mapping and on minimisation over gene-tree rootings. The extended-species-tree construction permits multi-copy comparisons but can be difficult to interpret for large copy-number imbalances. Hdist and the approximate SPR score require independent correctness and sensitivity benchmarks. Finally, rescaling does not remove all effects of gene-family size or taxon coverage, so these quantities must be measured as possible confounders rather than assumed to have been normalised away.

Gene-tree uncertainty is another unresolved dimension. Bootstrap or posterior trees can be projected as separate rows, but the current API does not prescribe whether a gene family should be represented by a mean signature, a distribution, or a hierarchical model. Collapsing uncertainty to one tree may create apparent clusters driven by inference error; treating every sampled topology as independent would instead exaggerate the effective sample size. A future extension should retain the grouping of topologies by family and propagate within-family uncertainty into downstream analyses.

The current results are therefore evidence of feasibility, not comparative superiority. The SPR-chain example is confounded by shared missingness within groups, the genome-level example is a single selected realisation, and the fungal partition lacks external biological validation. A publication-ready evaluation should use repeated simulations, predefined quantitative criteria, held-out families, simple nuisance-feature controls, and pairwise or quartet-based baselines where their input assumptions are satisfied. Empirical clusters should be tested for stability and enrichment against annotations that were not used to construct the signatures.

Subject to these validations, landmark representations offer a useful bridge between phylogenetic algorithms and general statistical learning. They allow multi-view feature selection, supervised prediction, and incremental analysis while keeping the phylogenetic meaning of the coordinates visible. The central claim is correspondingly modest: a fixed set of species-tree landmarks and complementary dissimilarities provides a workable common feature space for gene trees that cannot all be compared directly under standard same-leaf-set distances.

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